

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain the working principle of an RTD and also discuss the 3 and 4 wire method and draw the graph between Temperature and resistance of Platinum and Nickel.
- Q.24 Explain the working Principle of a Venturimeter with a neat Diagram. How is it used to Measure Flow Rate and Derive the Expression for Flow Rate Through a Venturimeter.
- Q.25 Explain one method of Low and High pressure measurement.

No. of Printed Pages : 4

221554

Roll No.

**5th Sem/
Instrumentation & Control
Subject : Process Instrumentation**

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 The unit of pressure is :
- a) Pascal
 - b) Newton
 - c) Bar
 - d) Both (a) and ©
- Q.2 In ultrasonic flow meters, the principle used is
- a) Bernoulli's theorem
 - b) Doppler effect or transit-time
 - c) Magnetic induction
 - d) Piezoelectric effect
- Q.3 RTDs work on which of the following principles
- a) See beck effect
 - b) Change in resistance with temperature
 - c) Change in voltage with current
 - d) Piezoelectric effect

- Q.4 The Bourdon tube is used to measure
- Temperature
 - Pressure
 - Flow
 - Humidity
- Q.5 What is the function of a signal conditioning circuit in a DAQ system.
- It digitizes the signal
 - It amplifies and filters the signal
 - It stores the signal
 - It converts signal to text
- Q.6 The output of a typical level transmitter is
- 0-10v
 - 4-20mA
 - 1-5V
 - 10-50mA

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Working Principle of Thermocouple_____.
- Q.8 Rotameter is Used to measure_____.
- Q.9 Thermistor is a type of (NTC) (Yes/No)
- Q.10 Which gauge is used to measure high pressure measurement_____.
- Q.11 Gamma rays method is used to level measurement. (Yes/No).
- Q.12 DAQ stands for_____.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Explain the working principle of the thermocouple and also discuss base and rare material of thermocouple.
- Q.14 Discuss the fiber optics thermometer.
- Q.15 What is pressure writes its units and how to measure pressure by using manometer.
- Q.16 Explain the different types of elastic pressure transducer.
- Q.17 Describe the working principle of a rotameter Derive the expression for pressure through Rotameter.
- Q.18 What is an ultrasonic flow motor and how to measure flow by using transit time method.
- Q.19 Explain the resistive method for Level Measurement.
- Q.20 Discuss the Gamma rays method for level measurement.
- Q.21 What is data acquisition and what are the main components of a DAQ system.
- Q.22 Explain the block diagram and Characteristics of Data logger